NBA Minutes Prediction

Game-Level Regression Model with Diagnostic Rigor

Problem Statement

Business Context: NBA coaches make rotation decisions based on patterns — player role, recent performance, team depth, opponent strength. Can we model these patterns?

Research Question: Given a player's role, recent performance, team context, and game conditions, how many minutes will they play?

Why This Matters: Minutes played is the foundation for all downstream predictions. You can't accurately predict points without predicting minutes first. This project teaches you to build a rigorous regression model on a real prediction problem.

Dataset & Features

Source: Basketball Reference (2024-25 or 2023-24 season)

Target Variable: Minutes Played (continuous, game-level)

13 Features:

Player-Level (Static): Position (G/F/C), Role (starter vs bench), Career avg minutes, Age
Team Context: Team pace, Team depth at position, Games played this season
Recent Performance (Last 10 Games): Avg minutes, Avg points, Avg fouls
Game Context: Opponent defensive rating, Home/Away, Rest days

Workflow by Phase

Phase 1: Data Collection & Exploration (3–4 hours)

Scrape Basketball Reference, EDA, visualize distributions, correlation matrix

Phase 2: Data Preparation (2–3 hours)

Handle missing values, engineer rolling averages, merge features

Phase 3: Linear Regression — Single Feature (2–3 hours)

Start simple: minutes ~ recent_avg_minutes, extract coefficients

Phase 4: Model Diagnostics & Assumptions (3–4 hours)

Normality, homoscedasticity, linearity, independence checks

Phase 5: Multiple Regression (2–3 hours)

Add all 13 features, compare to simple model

Phase 6: Inference & Interpretation (2–3 hours)

Confidence intervals, statistical significance, practical interpretation

Phase 7: Predictions & Visualization (2–3 hours)

Generate predictions, prediction intervals, visualize results

Phase 8: Refactor & Document (2–3 hours)

Move to src/ scripts, clean README

Total: ~19–26 hours

Learning Goals (Aligned to Coursework)

Modules 6–7: Pandas for data manipulation
Modules 8–10: Visualization (distributions, residuals, prediction intervals)
Module 11: Linear Regression Assumptions (normality, homoscedasticity, linearity)
Module 12: OLS fitting and interpretation
Module 13: Multiple regression
Module 14: Evaluation (R-squared, F-statistic, p-values, diagnostics)

Skills Developed

Skill	What You'll Learn
Web Scraping	Extract structured data from Basketball Reference
Feature Engineering	Rolling averages, categorical encoding, team-level joins
Statistical Thinking	Assumptions, diagnostics, inference from coefficients
Regression Rigor	OLS, R-squared, F-tests, p-values, confidence intervals
Visualization	Diagnostic plots, prediction ribbons, scatter + line
Python Structure	Modular scripts (not just notebooks), reusable functions
Problem Framing	Real prediction with interpretable features and practical meaning

Repo Structure

nba-minutes-prediction/
■■■ README.md
■■■ .gitignore
■■■ requirements.txt
■■■ data/
■ ■■■ nba_2024_25.csv
■■■ notebooks/
■ ■■■ exploration.ipynb
■■■ src/
■ ■■■ scrape.py
■ ■■■ prepare.py
■ ■■■ model.py
■ ■■■ visualize.py
■■■ outputs/
■■■ model_summary.txt
■■■ diagnostics_qq_plot.png
■■■ residuals_plot.png
■■■ predictions_vs_actual.png

Success Criteria

✓ Successfully scrape one season of NBA data (100+ games, 50+ players)

✓ Feature engineering complete (13 features + target)

✓ Linear regression model fits with interpretable coefficients

✓ Diagnostic plots show model assumptions (or clearly document violations)

✓ Multiple regression compares to simple model with justification

✓ Prediction intervals visualized cleanly

✓ Refactored into modular src/ scripts

✓ README explains problem, data, findings, limitations

Next Steps (After This Project)

Once you complete this, you'll have hands-on experience with the full regression workflow. You can then evaluate whether to deepen sports methodology or pivot to fintech methodology, applying the same rigor to credit and fraud problems.

Ready to Start Tomorrow

You're ready. Scope is clear, features are defined, workflow is mapped to your coursework. Build it clean, document as you go.